

R.O. 11/05/2021

$$\min 2x_1 + x_2 - x_3$$

$$\bar{x} = \begin{pmatrix} 4 \\ 2/3 \\ 0 \end{pmatrix}$$

$$\begin{cases} x_1 + 3x_2 \leq 6 \\ 3x_1 - x_2 + 4x_3 \geq 8 \end{cases}$$

v_1

v_2

$$m=2$$

$$n=5$$

vincoli

variabili

constraint slack

$$x_1, x_2 \geq 0 \quad x_3 \leq 0$$

a) $\bar{x} = \begin{pmatrix} 4 \\ 0 \\ 0 \end{pmatrix}$ E' UNA SOLUZIONE DI BASE AMMISSIBILE S.B.A.?

$\bar{x}$ E' ammissibile?

$$\begin{matrix} 4 \leq 6 \\ 12 \geq 8 \end{matrix} \quad \text{si}$$

$$4 \leq 6$$

$$12 \geq 8$$

$$s_1 = 2$$

$$s_2 = 4$$

$$x_1 + 3x_2 + s_1 = 6$$

$$3x_1 - x_2 + 4x_3 - s_2 = 8$$

$$\bar{x} = \begin{pmatrix} 4 \\ 0 \\ 0 \\ 2 \\ 4 \end{pmatrix}$$

NON E' UNA
SOLUZIONE DI BASE
AMMISSIBILE

b) $\exists$ vertice $\bar{x}$ con componenti x_1 e x_2 ; $x_1 > 0$?
 $x_2 > 0$?



VERTICE $\Rightarrow$ S.B.A

$$\Rightarrow x_3 = 0$$

$$s_1 = 0$$

$$s_2 = 0$$

$$x_1 + 3x_2 = 6$$

$$3x_1 - x_2 = 8$$

$$x_2 = -3x_1 + 6$$

$$x_1 - 9x_1 + 24 = 6$$

$$x_1 = 3 > 0$$

$$x_2 = 1 > 0$$

Si esiste
un vertice con
 $x_1 > 0, x_2 > 0$

c) Può esistere una soluzione ottima con x_3 in base?

$$x_3 < 0 \quad x_3 \neq 0$$

$x^*, x_3 \neq 0 \Rightarrow$ scarto duale del terzo vincolo (Problema Duale) $\neq$ nullo

$$\sum x_i s d_i = 0$$

$$\max \quad 6y_1 + 8y_2$$

$$\text{s.v.} \quad y_1 + 3y_2 \leq 2 \quad \text{VD 1}$$

$$3y_1 - y_2 \leq 1 \quad \text{VD 2}$$

$$4y_2 \geq -1 \quad \text{VD 3} \quad x_3 \neq 0$$

$$y_1 \leq 0$$

$$y_2 \geq 0$$

NO!

$$4y_2 = -1$$

$\Rightarrow$

$$y_2 = -\frac{1}{4}$$

non esiste
una soluz.
ottima del
primale con $x_3 \neq 0$

d) APPLICARE IL SIMPLESSO DUALE AL PROBLEMA a)

Forma STANDARD:

$$\min \quad 2x_1 + x_2 + x_3$$

$$\text{s.v.} \quad x_1 + 3x_2 + x_4 = 6$$

$$3x_1 - x_2 - 4x_3 - x_5 = 8$$

$$x_1, x_2, x_3, x_4, x_5 \geq 0$$

$$\begin{array}{c|ccccc}
 & x_1 & & & & \\
 0 & 2 & 1 & 1 & 0 & 0 \\
 6 & 1 & 3 & 0 & 1 & 0 \\
 \rightarrow -8 & -3 & 1 & 4 & 0 & 1
 \end{array}$$

Pivot(2,1)

$$\begin{array}{c|ccccc}
 & x_1 & x_2 & x_3 & x_4 & x_5 \\
 -16/3 & 0 & -5/3 & 11/3 & 0 & 2/3 \\
 x_4 \cdot 10/3 & 0 & 4/3 & 4/3 & 1 & 1/3 \\
 x_1 \cdot 8/3 & 1 & -1/3 & -4/3 & 0 & -1/3
 \end{array}$$

$0 \leq 7, 0$ Ammissibilità Dual

$0 \leq 7, 0$ Ammissibilità Primal

$$Z^* = 16/3$$

$$x^* = (8/3, 0, 0, 10/3, 0)$$

e) Eseguire la prima iterazione dell'algoritmo del primal-dual partendo dalla soluz. $y^0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$$\text{Max } 6x_1 + 8x_2$$

$$x_1 + 3x_2 \leq 2$$

$$3x_1 - x_2 \leq 1$$

$$-4x_2 \leq 1$$

$$x_1 \leq 0$$

$$-x_2 \leq 0$$

$$y^{(1)} = y^0 + \theta \pi^0$$

$$y^0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\langle \bar{y}, sd \rangle = 0$$

$$0 \leq 2 \quad sd_1 = 2 \Rightarrow x_1 = 0$$

$$0 \leq 1 \quad sd_2 = 1 \Rightarrow x_2 = 0$$

$$0 \leq 1 \quad sd_3 = 1 \Rightarrow x_3 = 0$$

$$0 \leq 0 \quad sd_4 = 0$$

$$0 \leq 0 \quad sd_5 = 0$$

P.R.

$$x_4 = 6$$

$$s_1 - x_5 = 8$$

$$\text{Min } s_1$$

$$x_4 = 6$$

P.R.

$$s_1 - x_5 = 8$$

$$\begin{array}{c|ccc} & x_4 & x_5 & s_1 \\ \hline 0 & 0 & 6 & 1 \\ 6 & 1 & 0 & 0 \\ 8 & 0 & -1 & 1 \end{array}$$

$$\begin{array}{c|ccc} & x_4 & x_5 & s_1 \\ \hline -8 & 0 & 1 & 0 \\ x_4 = 6 & 1 & 0 & 0 \\ s_1 = 8 & 0 & -1 & 1 \end{array}$$

$$z^* = 8 > 0$$

$\Rightarrow \bar{x}$ non Ammiss.

$$\bar{x} = (0, 0, 0, 1, 0)$$

$x_4 \quad x_5$

Duale del P.R. (D.R.)

$$\text{Max} \quad 6\pi_1 + 8\pi_2$$

$$x_4 > 0 \quad \pi_1 \leq 0 \quad \Rightarrow \quad \pi_1^* = 0$$

$$x_5 = 0 \quad -\pi_2 \leq 0$$

$$s_1 > 0 \quad \pi_2 \leq 1 \quad \Rightarrow \quad \pi_2^* = 1 \quad \pi^* = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$A^T \pi^* = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3b_2 \leq 2 \\ -b_2 \leq 1 \\ -4b_2 \leq 1 \\ 0 \leq 0 \\ -b_2 \leq 0 \end{bmatrix}$$

$$\begin{aligned} 0 &\leq \frac{2}{3} \\ b_2 &\geq -1 \\ b_2 &\geq -\frac{1}{4} \end{aligned}$$

$$b_2 \geq 0 \quad 0 \leq b_2 \leq \frac{2}{3}$$

$$\Rightarrow \theta_c^* = \left(\frac{2}{3}\right)$$

$$y^{(1)} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + \frac{2}{3} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 2/3 \end{pmatrix}$$

FINE
ESERCIZIO.

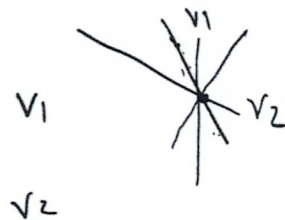
ESERCIZIO 2

$$\max b = 2x_1 - 3x_2 - x_3$$

$$x_1 + 2x_2 \leq 6$$

$$x_1 - x_2 + 3x_3 \geq 6$$

$$x_1, x_2, x_3 \geq 0$$



2a) verificare se i seguenti punti siano o meno
S.B.A.

$$\bar{x}_1 = \begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix} \quad \bar{x}_2 = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} \quad \bar{x}_3 = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$$

$$6 = 6$$

$$s_1 = 0$$

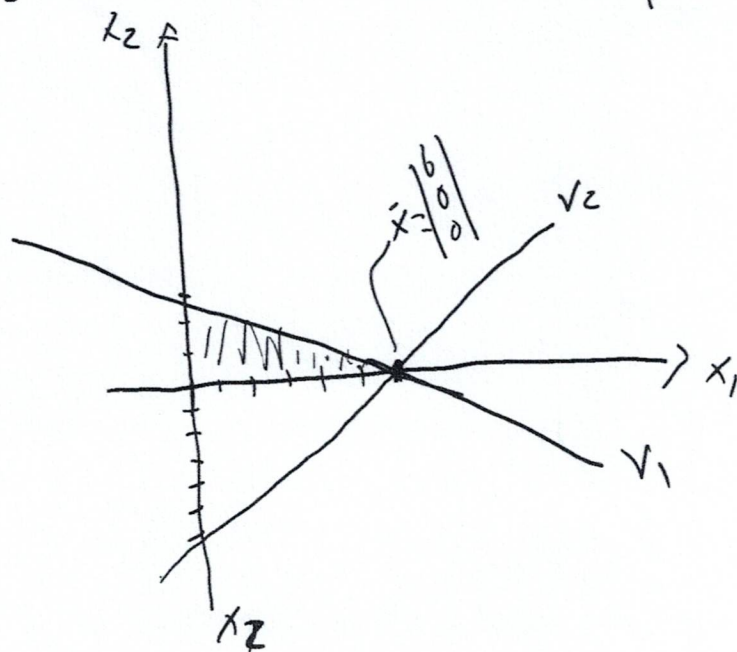
$$6 = 6$$

$$s_2 = 0$$

$$\bar{x}_1 = \begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix}$$

è JNA
? S.B.A.
DEGENERE!!

$$x_3 = 0$$



$$x_1 \geq 0$$

		x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	s_1
	$-25/3$	0	$4/3$	$19/6$	$9/2$	0	0	0	$7/6$	0
x_5	1	0	1	$-1/2$	$-3/2$	1	0	0	$3/2$	0
x_1	$11/3$	1	$-2/3$	$-1/3$	0	0	0	0	$2/3$	0
x_6	$2/3$	0	$1/3$	$1/6$	$-1/2$	0	1	0	$7/6$	0
x_7	1	0	-3	$1/2$	$9/2$	0	0	1	$-15/2$	0
		0	0	0	0	0	0	0	0	1
	(3)	1								

$x^* = (11/3, 0, 0, 0, 1, 2/3, 1)$

$-2/3$

$$x_1 \leq \lfloor 11/3 \rfloor = 3$$

$$x_3 + s_3 = 3$$

riga 5 - riga 2

$$r_5 \quad -2/3 \mid 0 \quad 2/3 \quad 1/3 \quad 0 \quad 0 \quad 0 \quad 0 \quad -2/3 \quad 1$$

Abbiamo un tableau con un $c > 0$
 $b: b_5 < 0$

$\Rightarrow$ APPLICHIAMO IL SIMPLESSO DUALE

